

Sum and Difference Formulas (6.3)

Name _____

Circuit

Begin with the problem in the top left. Answer the question. To advance in the circuit, find your answer and write “2” in the blank. Solve that problem, find your answer and write “3” in the blank and so on until you have solved every problem and returned to the beginning.

ANSWER: $\frac{\sqrt{2} - \sqrt{6}}{4}$

_____ Find the exact value of $\cos 15^\circ$.

ANSWER: $\frac{1}{2}$

_____ Find the exact value of $\tan 75^\circ$.

ANSWER: $\frac{\sqrt{6} - \sqrt{2}}{4}$

_____ Find the exact value of $\tan 15^\circ$.

ANSWER: $\sin x$

_____ Simplify $\sin\left(\frac{\pi}{2} - \theta\right)$.

ANSWER: $\frac{\sqrt{6} + \sqrt{2}}{4}$ # _____ Find the exact value of $\sin 15^\circ$.	ANSWER: $-\sin x$ # _____ Simplify $\cos\left(x - \frac{\pi}{2}\right)$.
ANSWER: $\frac{\sqrt{3}}{2}$ # _____ Simplify $\cos 70^\circ \cos 10^\circ + \sin 70^\circ \sin 10^\circ$.	ANSWER: $\cos \theta$ # _____ Find the exact value of $\cos 105^\circ$.
ANSWER: $2 - \sqrt{3}$ # _____ Simplify $\sin 40^\circ \cos 20^\circ + \cos 40^\circ \sin 20^\circ$.	ANSWER: $2 + \sqrt{3}$ # _____ Simplify $\sin(x + \pi)$.